Pythia training dynamics: Adam adaptive directions

Pythia-14M Pythia-31M Pythia-70M

Updates 1-100 | N=5

Updates 100-400 | N=21

Change in ln(training loss)

Projected direction change $q_t^{(d)}$

Adam-direction cosine $\rho_t^{(d)}$

Adam direction $\|d_t\|_2$

Optimizer update

Optimizer update

$d_t = \widehat{m}_t/(\sqrt{\widehat{v}_t} + \varepsilon)$; $q_t^{(d)} = 1 - \rho_t^{(d)} \|d_t\|/\|d_{t-1}\|$ (direction change, not measured curvature).

Complete trailing means: N=5 left, N=21 right. Loss uses $\ln L_t - \ln L_{t-1}$. Seed 1235. LR and decay excluded from d.

Y limits vary by column; norm row uses log Y. Update 100 appears twice; smoothing spans the boundary.

Left means start at update 5 (norm) / 6 (other rows). Direction cosines use a narrow Y range; all three are positive here.